

Assignment JC5

Handong Honor Code

- It is your responsibility to understand and adhere to the Handong Honor Code.
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Goals

- To practice creating and managing AWS cloud resources such as S3 buckets, IAM roles, and EC2 instances.
- To understand how to configure secure access between AWS services using IAM roles and permissions.
- To develop skills in uploading, storing, and transferring files between local machines, Amazon S3, and EC2 instances.
- To gain hands-on experience using AWS CLI commands for cloud resource management and file operations.
- To learn how to set up a Python data analysis environment on an EC2 instance using Miniconda and JupyterLab.
- To understand the workflow of cloud-based data processing and notebook management in AWS.

IMPORTANT NOTES

Ensure that you assign the **Owner** tag to all the cloud resources you create, with the value set to your *student ID (학번)*.

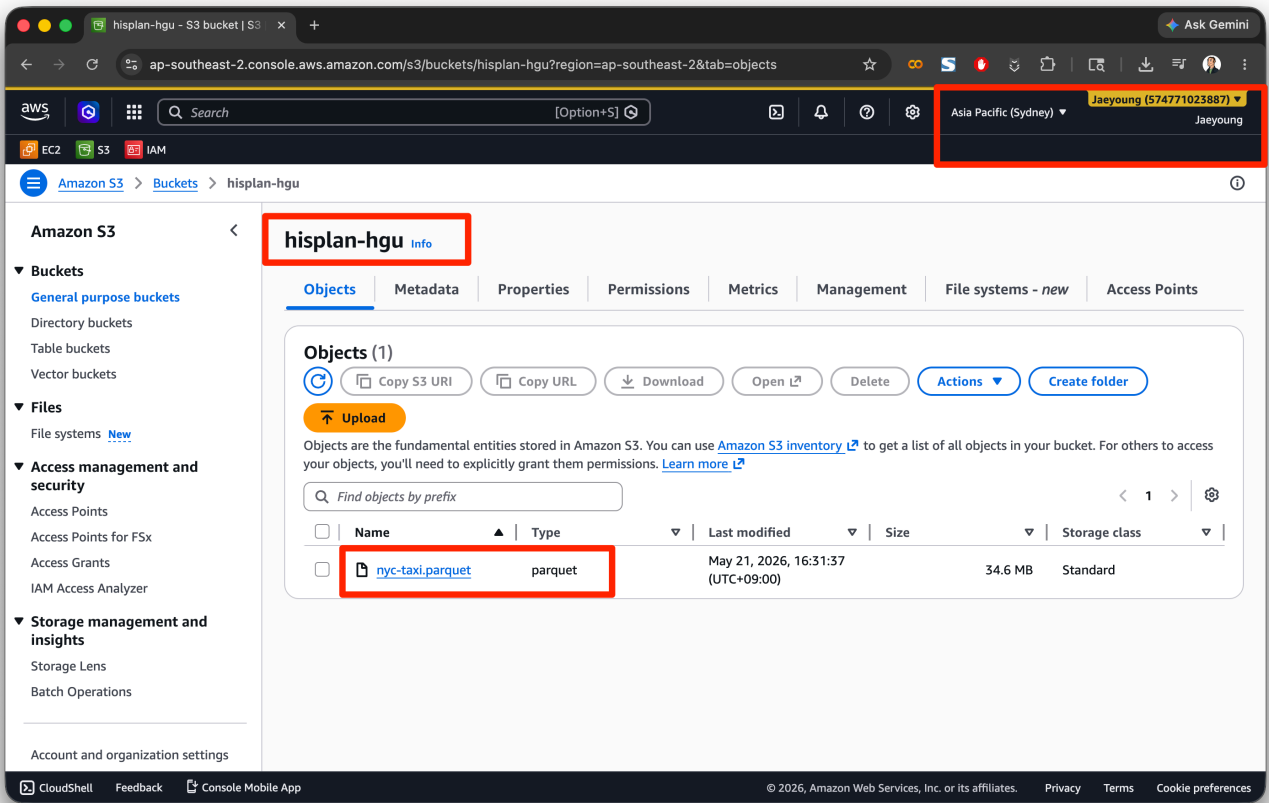
You will be asked to take a screenshot for each task. When taking the screenshots, there are a couple of important points to keep in mind:

1. Ensure to include the entire screen in the capture, not just the relevant section. This is to ensure that we can verify the context of the screenshot and confirm that it was taken by you.
2. You will be provided with an example of how the screenshot should look like for each task. Please follow the format shown in the examples when taking your screenshots, especially if there is a *red box* or *red arrow* highlighting a specific area. The red box or arrow is meant to indicate the specific section that we want you to focus on, but the entire screen should be visible in the screenshot.

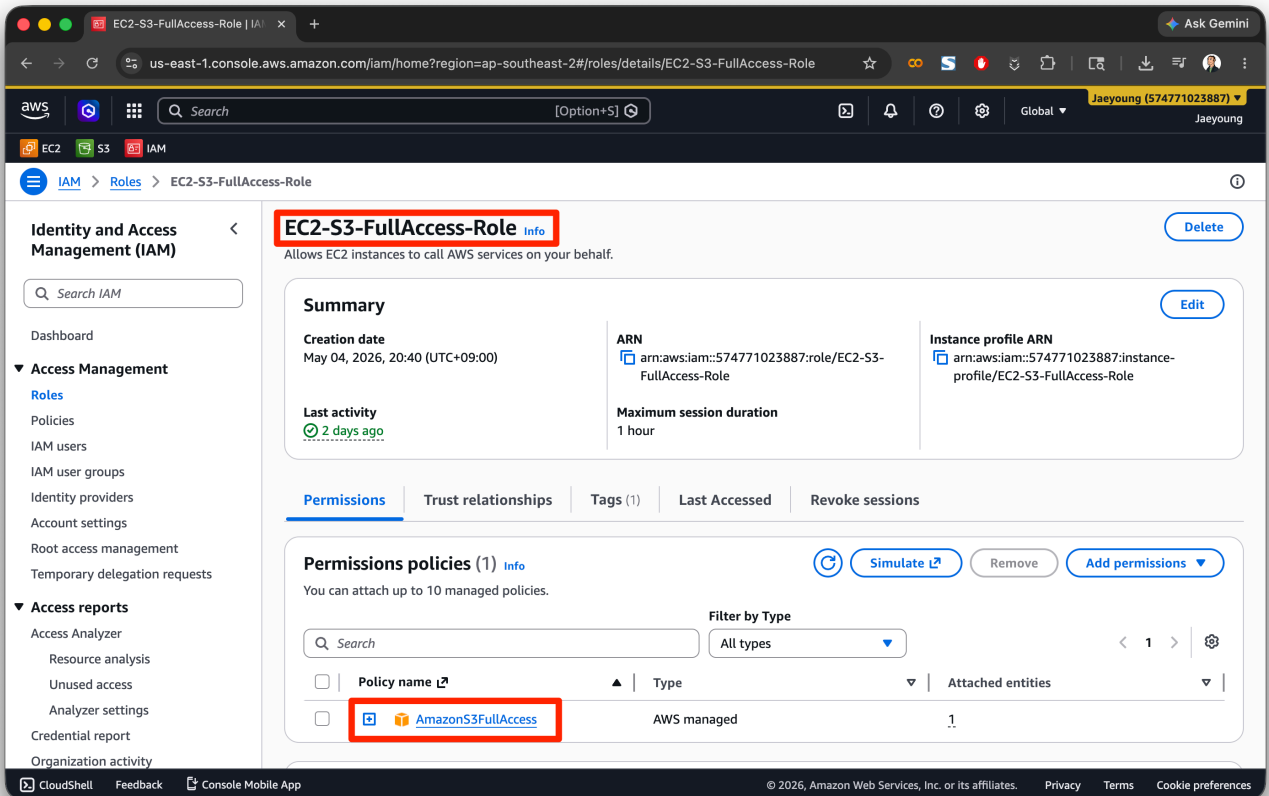
1. Create an S3 Bucket

1. Create an S3 bucket with the name **hgu-22316316** (replace **22316316** with your student ID).
2. Download **nyc-taxi.parquet** from LMS to your local machine.
3. Upload the **nyc-taxi.parquet** file from your local machine to your S3 bucket.

4. Take a screenshot and save it as `s3-bucket.png`.



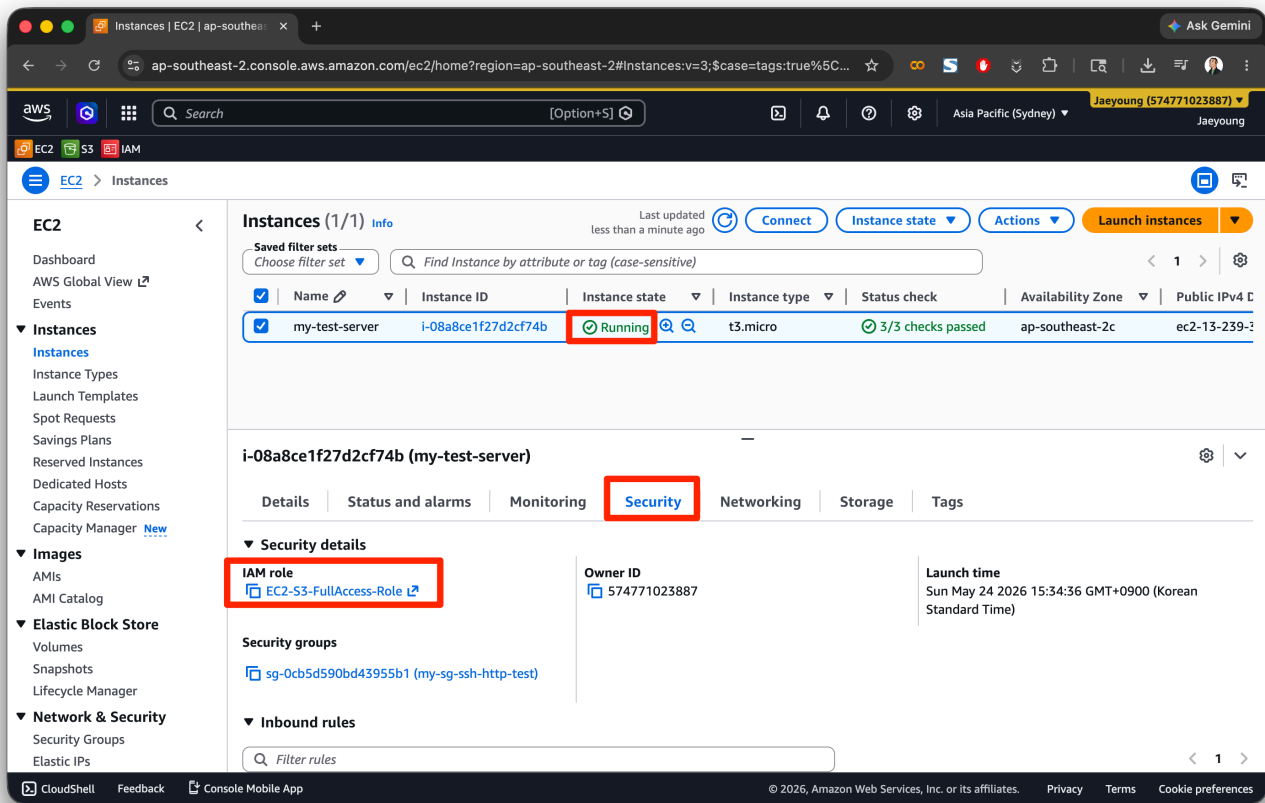
2. Create an IAM Role with S3 Full Access



1. Create an IAM role with the name **EC2-S3-FullAccess-Role** and attach the **AmazonS3FullAccess** policy to it.
2. Take a screenshot of the IAM role details and save it as **iam-role.png**.

3. Launch an EC2 Instance with the IAM Role

1. Launch an EC2 instance with IAM role **EC2-S3-FullAccess-Role** attached to it.
2. Once the instance is running, select the **Security** tab in the EC2 dashboard.
3. Take a screenshot showing the attached IAM role. Save the screenshot as **ec2-iam-role.png**.

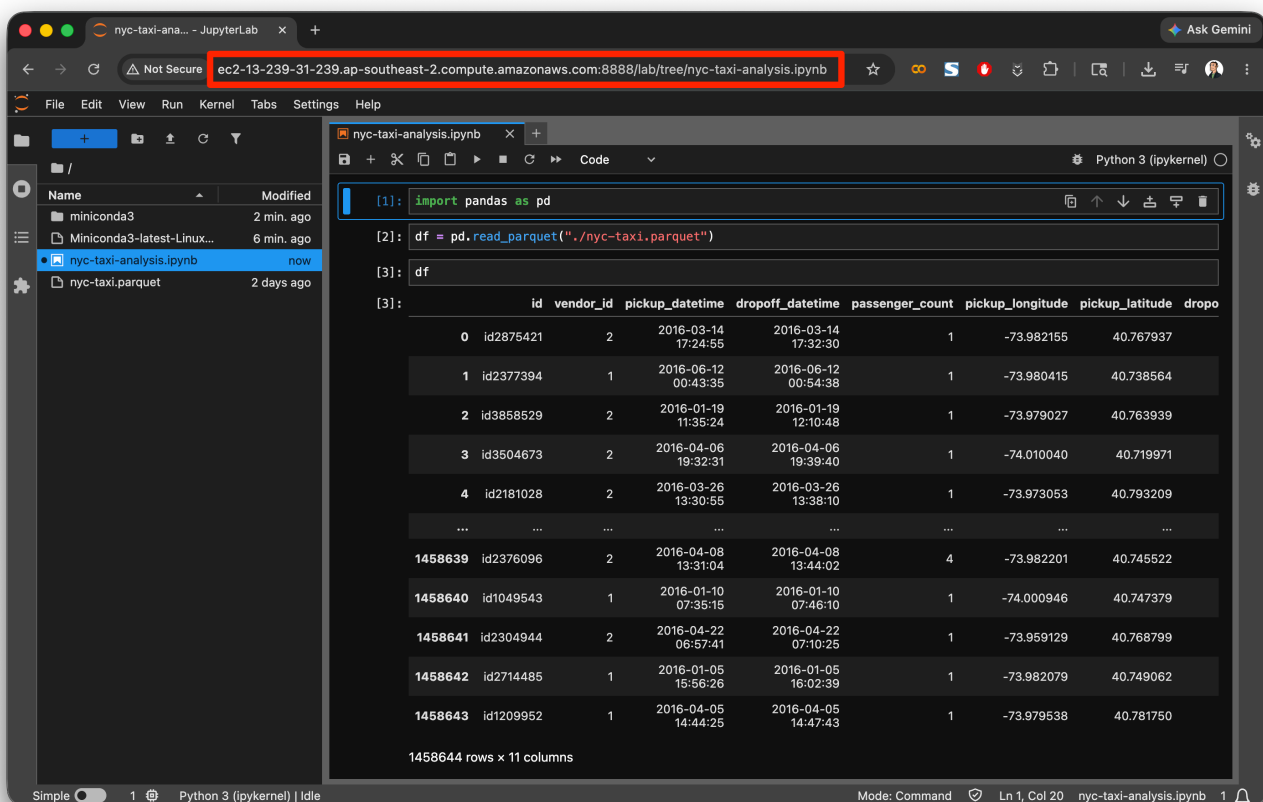


4. Copy from S3 to EC2

1. Connect to your EC2 instance using SSH.
2. Use the AWS CLI to copy the **nyc-taxi.parquet** file from your S3 bucket to the EC2 instance.
3. Take a screenshot of the terminal showing the commands used to copy the file and the successful completion message. Save the screenshot as **ec2-terminal-s3.png**.

```
ec2-user@ip-172-31-31-166:~  
[ec2-user@ip-172-31-31-166 ~]$ aws s3 ls s3://hisplan-hgu/  
2026-05-21 07:31:37    36294724 nyc-taxi.parquet  
[ec2-user@ip-172-31-31-166 ~]$ aws s3 cp s3://hisplan-hgu/nyc-taxi.parquet .  
download: s3://hisplan-hgu/nyc-taxi.parquet to ./nyc-taxi.parquet  
[ec2-user@ip-172-31-31-166 ~]$ ls -l  
total 35448  
-rw-r--r--. 1 ec2-user ec2-user 36294724 May 21 07:31 nyc-taxi.parquet  
[ec2-user@ip-172-31-31-166 ~]$
```

5. Read Parquet File in JupyterLab



The screenshot shows the JupyterLab interface. The left sidebar displays the file explorer with the following items:

Name	Modified
miniconda3	2 min. ago
Miniconda3-latest-Linux...	6 min. ago
nyc-taxi-analysis.ipynb	now
nyc-taxi.parquet	2 days ago

The main area shows the Jupyter notebook 'nyc-taxi-analysis.ipynb' with the following code cells:

```
[1]: import pandas as pd  
[2]: df = pd.read_parquet("./nyc-taxi.parquet")  
[3]: df
```

The output of the third cell is a DataFrame with 1458644 rows and 11 columns. The columns are: id, vendor_id, pickup_datetime, dropoff_datetime, passenger_count, pickup_longitude, pickup_latitude, and dropoff_latitude. The output shows a preview of the data:

	id	vendor_id	pickup_datetime	dropoff_datetime	passenger_count	pickup_longitude	pickup_latitude	dropoff_latitude
0	id2875421	2	2016-03-14 17:24:55	2016-03-14 17:32:30	1	-73.982155	40.767937	
1	id2377394	1	2016-06-12 00:43:35	2016-06-12 00:54:38	1	-73.980415	40.738564	
2	id3858529	2	2016-01-19 11:35:24	2016-01-19 12:10:48	1	-73.979027	40.763939	
3	id3504673	2	2016-04-06 19:32:31	2016-04-06 19:39:40	1	-74.010040	40.719971	
4	id2181028	2	2016-03-26 13:30:55	2016-03-26 13:38:10	1	-73.973053	40.793209	
...	...	...	...	...	...	...	...	...
1458639	id2376096	2	2016-04-08 13:31:04	2016-04-08 13:44:02	4	-73.982201	40.745522	
1458640	id1049543	1	2016-01-10 07:35:15	2016-01-10 07:46:10	1	-74.000946	40.747379	
1458641	id2304944	2	2016-04-22 06:57:41	2016-04-22 07:10:25	1	-73.959129	40.768799	
1458642	id2714485	1	2016-01-05 15:56:26	2016-01-05 16:02:39	1	-73.982079	40.749062	
1458643	id1209952	1	2016-04-05 14:44:25	2016-04-05 14:47:43	1	-73.979538	40.781750	

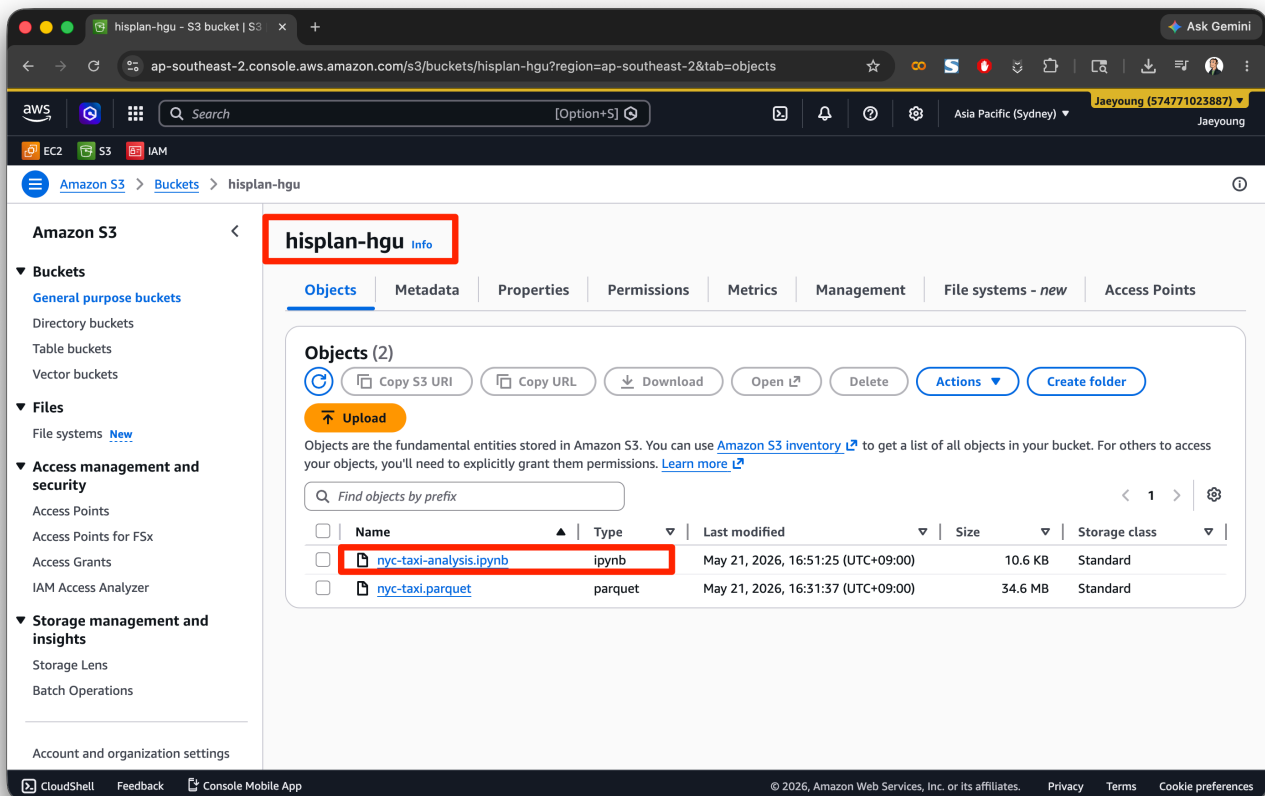
The status bar at the bottom indicates 'Simple', 'Python 3 (ipykernel)', and 'Mode: Command'.

1. Install Miniconda.
2. Create a new Conda environment named `dev`, and activate it.
3. Install Python, Pip, JupyterLab, and the necessary libraries to read Parquet files on your EC2 instance.
4. Start JupyterLab on your EC2 instance.

5. Open a new notebook in JupyterLab and write code to read and display the trip data from the `nyc-taxi.parquet` file.
6. Save the notebook as `nyc-taxi-analysis.ipynb`.
7. Take a screenshot of the JupyterLab notebook showing the code and the output of reading the Parquet file. Ensure that the IP address of your EC2 instance is visible in the screenshot. Save the screenshot as `jupyterlab-parquet.png`.

6. Download the Notebook to Local Machine

1. Use AWS CLI to securely copy the `nyc-taxi-analysis.ipynb` notebook from your EC2 instance to your S3 bucket.
2. Use AWS CLI to securely copy the `nyc-taxi-analysis.ipynb` notebook from your S3 bucket to your local machine.
3. Take a screenshot of the S3 bucket showing the notebook file. Save the screenshot as `s3-ipynb.png`.



Submission

When submitting, please package all the files into a single ZIP archive named `학번.zip` and submit it.

Example of submission format:

```
22316316.zip
├─ nyc-taxi-analysis.ipynb
├─ s3-bucket.png
└─ iam-role.png
```

- └─ ec2-iam-role.png
- └─ ec2-terminal-s3.png
- └─ jupyterlab-parquet.png
- └─ s3-ipynb.png